

Central Bank Digital Currency, Monetary Policy, and Macroeconomy : Central Bank Digital Currency, Monetary Policy, and Macroeconomy

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CENTRAL BANK DIGITAL CURRENCY, MONETARY POLICY, AND MACROECONOMY: EVIDENCE FROM INDONESIA

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ABSTRACT

This paper studies the impact of a universally accessible interest-bearing Central Bank Digital Currency (CBDC) on macroeconomic stability and monetary policy in Indonesia. We construct a simple reduced-form New Keynesian model calibrated with monthly data for Indonesia from July 2005 to February 2022. We find that the CBDC regime significantly improves macroeconomic stability in Indonesia. In this regard, the CBDC rate, as an additional monetary policy instrument, responsively reacts to macroeconomic fluctuations and significantly impacts the demand side. Our results also suggest that monetary policy is more effective under the CBDC regime than under the non-CBDC.

Keywords: Central bank digital currency; Monetary policy; Macroeconomy; Indonesia.

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I. INTRODUCTION

The governor of the Central Bank of Indonesia (Bank Indonesia hereafter), Perry Warjiyo, has made a public statement about the intention of Bank Indonesia to issue Central Bank Digital Currency (CBDC hereafter) in the future (Reuters, 2021). Bank Indonesia is currently organizing internal research to address this issue. With an intention to accelerate the issuance of CBDC, Bank Indonesia now prioritizes the digital payment advancement policy, supporting payment system digitalization. Such an intention is reflected by, first, the issuance of QR Code Indonesian Standard (QRIS) as a uniform digital-based transaction and BI-Fast to facilitate costless digital-based banking transactions. Second, Bank Indonesia has also initiated the National Working Group to Accelerate and Expand Local Digitalization (P2DD). This working group specifically encourages the national and local level government to be proactive in promoting digital-based transactions in the local area. Accordingly, a two-digit growth rate characterizes digital transactions in Indonesia over the last few years. In May 2021, Bank Indonesia documented a 57.38 percent year-on-year growth in electronic money transactions worth 23.7 trillion transaction increases. In the digital banking sector, the volume and value of transactions also increased significantly by 56.49 percent and 66.41 percent, respectively (Bank Indonesia, 2021). In the second quarter of 2022, the growth of digital transactions in Indonesia remains high, with a 39.85 percent year-on-year growth in electronic money transactions and a 38.45 percent increase in digital banking transactions (Bank Indonesia, 2022).

Concerns about the issuance of CBDC in Indonesia emerge, we argue, three fascinating questions, especially if the CBDC is universally accessible and interest-bearing.¹ The first question that appears is related to whether or not a universally accessible interest-bearing CBDC supports Bank Indonesia's objective to manage macroeconomic stability. This question is central to be addressed in the case of Indonesia as Juhro (2021) specifically mentions that Bank Indonesia always pays attention to the potential implications of issuing CBDC on macroeconomic stability. Furthermore, we ask how the interest rate embedded in CBDC, as a policy instrument, will respond to and play a role in stabilizing macroeconomic fluctuations. Bordo and Levin (2017) explain that the interest rate binding to the CBDC provides a secure store of value of money and can serve as the main tool for conducting monetary policy. In this case, the CBDC rate could act as a business cycle stabilizer through which the central bank conducts the price-based CBDC policy in a countercyclical manner (Barrdear and Kumhof, 2021; Bindseil, 2020; Meaning *et al.*, 2021). Lastly, we also look into whether the existence of CBDC enhances the effectiveness of the monetary policy. Mancini-Griffoli *et al.* (2018) suggest that the interest rate channel may be the most affected and could be strengthened by the presence of a universally accessible interest-bearing CBDC because changes in interest rate directly affect every economic agent.

¹ The term CBDC belongs to several concepts. The first concept, the so-called universally accessible CBDC, proposes a widely accessible CBDC to the end-users. The second one, so-called wholesale CBDC, suggests a narrowly provided CBDC only to particular institutions such as commercial banks and non-bank financial institutions. Furthermore, both types of CBDC, universally accessible and wholesale, can also be interest-bearing or not (Bech and Garratt, 2017).

Any study elaborating on the issues of CBDC in the context of Indonesia, unfortunately, has not formally addressed the macroeconomic stability and monetary policy implication due to the issuance of CBDC. Harahap *et al.* (2017) explore the effect of CBDC on monetary policy and macroeconomy by emphasizing the financial technology improvement due to the CBDC. By using the Computational General Equilibrium (CGE) simulation, they find that the issuance of CBDC will initially increase the deposit rate, increases national and regional economic growth, and increases the employment rate. However, none of their findings address the macroeconomic stability impact of the CBDC in Indonesia. Purnawan and Riyanti (2019) elaborate on the effect of CBDC on the central bank's policy design qualitatively and argue that CBDC provides several benefits such as a new monetary policy instrument, increasing financial inclusion, and reducing transaction costs. More recently, Zams *et al.* (2020) analyze the optimal CBDC design for Indonesia by collecting experts' opinions. They argue that, based on Analytical Network Process (ANP) analysis, non-interest-bearing CBDC is the most ideal type for Indonesia.

In this paper, we attempt to investigate the implication of a universally accessible interest-bearing CBDC on macroeconomic stability and monetary policy in Indonesia. To this end, we first analyze the magnitude of macroeconomic fluctuations in response to the demand and supply shocks under the CBDC regime and non-CBDC regime. Second, we scrutinize the role of the interest rate embedded in CBDC as a policy instrument in response to macroeconomic shocks. Third, we assess and compare the effectiveness of monetary policy in the CBDC regime and the non-CBDC regime. To accommodate our objectives, we construct a simple reduced-form New Keynesian (NK) model calibrated to the monthly data for Indonesia from July 2005 to February 2022. We characterize the existence of a universally accessible interest-bearing CBDC by introducing the CBDC policy rate which is conducted separately from the conventional policy rate. In this regard, we specify the CBDC rate reaction function using a countercyclical rule following Barrdear and Kumhof (2021).

We conjecture that the presence of a universally accessible interest-bearing CBDC enhances the macroeconomic stability and monetary policy effectiveness in Indonesia. The transmission is quite straightforward. Changes in the interest rate embedded in the CBDC influence the income of its holders. The higher (lower) interest rate on CBDC increases the return of holding CBDC and reduces transaction costs. For households, this increases (decreases) household income and increases (decreases) the ability to consume. Consequently, a higher (lower) interest rate on CBDC makes CBDC more (less) attractive. It in turn injects (withdraws) liquidity into the economy, expanding (contracting) the demand-side economic activity (Barrdear and Kumhof, 2021; Bordo and Levin, 2017; Gross and Schiller, 2020; Nebbi *et al.*, 2022). Furthermore, the effectiveness of the monetary policy, we argue, will improve under the CBDC regime. This is because the CBDC rate directly affects every economic agent not only the banking sector but also households and firms. This makes the impact of changes in the CBDC rate more directly connected to the real sector under the CBDC regime (Copic and Franke, 2020; Nebbi *et al.*, 2022).

This paper contributes to the current literature by highlighting fresh insights in the context of Indonesia. We find that the CBDC regime significantly improves macroeconomic stability in Indonesia. Our result indicates that macroeconomic shocks are significantly less volatile and more quickly stabilized under the CBDC regime than in the non-CBDC regime. In this regard, the CBDC rate, as an additional monetary policy instrument, responsively reacts to macroeconomic fluctuations and significantly impacts the demand side. Our results also suggest that the sacrifice ratio, as the proxy of monetary policy effectiveness, produced under the CBDC regime due to discretionary policy shocks is significantly lower than in the non-CBDC regime – meaning that the output loss due to a reduction in the inflation rate is relatively small under the CBDC regime. This indicates that monetary policy is more effective under the CBDC regime than under the non-CBDC.

As our objective is only to investigate the macroeconomic stability implications of CBDC issuance in Indonesia, we argue that a simple macroeconomic model is sufficient. However, there is a lot of room for exploration that this paper has not mentioned in detail. First, this paper addresses the heterogeneous household preferences in money holding and the network effect. By addressing this aspect, we acknowledge that it would deliver a richer analysis regarding the magnitude of the impact of issuing a universally accessible interest-bearing CBDC. For instance, Agur, Ari, and Dell’Ariccia (2022) demonstrate that the amount of CBDC held by the household may be heterogeneous which is dependent on the degree of anonymity preference. Also, the network effect plays an important role in determining the degree of CBDC useability and, therefore, its consequences on the economy. Second, this paper addresses the explicit aspect of the market structure of the banking sector. As is illustrated by numerous works in the CBDC-related literature, the structure of the banking sector may also play an important role in the transmission mechanism of the CBDC rate (Andolfatto, 2018; Fernández-Villaverde, Sanches, Schilling, and Uhlig, 2020; Keister and Sanches, 2019; Kim and Kwon, 2019).

The rest of this paper is organized as follows. Section II elaborates on the literature review. Section III demonstrates extending the NK model. Section IV shows the data profile and calibration parameters. Sections V and VI elaborate on the findings and robustness tests. Section VII provides concluding remarks.

II. LITERATURE REVIEW

Discourse on CBDC has drawn significant attention from central bankers. Such attention has been primarily driven by the potential enhancement of monetary policy, financial stability, financial inclusion, and payment efficiency. A survey conducted by Codruta and Wehrli (2021) illustrates that 52 out of 60 observed central banks are now exploring the benefits and drawbacks of the CBDC issuance, conducting experimentation, and implementing the CBDC pilot project.

The term CBDC belongs to several concepts but in general Meaning *et al.* (2021) formally define CBDC as “any electronic, fiat liability of a central bank that can be used to settle payments, or as a store of value.” This implies that CBDC virtually already exists in the current system, namely central bank reserves. More specifically, Bech

and Garratt (2017) also develop the taxonomy of the CBDC which is divided into two general categories. The first concept, the so-called universally accessible CBDC, proposes a widely accessible CBDC to the end-users. The second one, the so-called wholesale CBDC, suggests a narrowly provided CBDC only to particular institutions such as commercial banks and non-bank financial institutions. Furthermore, both types of CBDC, universally accessible and wholesale, can also be interest-bearing or not. Nowadays, wholesale CBDC is widely considered to be issued in advanced economies, while emerging economies are more focused on developing universally accessible CBDC (Maryaningsih, Nazara, Kacaribu, and Juhro, 2022).

In this regard, a universally accessible interest-bearing CBDC is certainly the most conceptually fascinating, especially for developing and emerging countries. This is because a universally accessible interest-bearing CBDC allows the citizens to access the central bank directly (Bech and Garratt, 2017; Fernández-Villaverde *et al.*, 2020; Kiff *et al.*, 2020). This assembles a very different circumstance from the current system and delivers substantial macroeconomic implications. On the other hand, wholesale CBDC relatively brings no significant consequences on monetary policy and financial stability. Bindseil (2020) points out that wholesale CBDC would not change the size of the monetary base so that it has no significant practical differences from the conventional system.

A growing number of literature has drawn significant concern on the macroeconomic implications of CBDC. Chen and Siklos (2022) argue that the CBDC will have a significant impact on the inflation rate since, principally, its introduction raises the speed and frequency of cash-like transactions. In this regard, they use McCallum's monetary rule framework and illustrate a robust finding that the presence of CBDC will not lead to loss of inflation control, but otherwise. Syarifuddin and Bakhtiar (2022) demonstrate that the presence of CBDC provides several macroeconomic benefits. More specifically, they find that CBDC offers a lower transaction cost that encourages household consumption and a steady path leading to lower inflation. Barrdear and Kumhof (2021) investigate the consequences of the introduction of interest-bearing CBDC using the United States (US) data before 2008 and find that it significantly contributes to macroeconomic stability, long-run economic growth, and inflation rate. More specifically, they find that the interest-bearing CBDC issuance equivalent to 30 percent of the US GDP increases the long-run GDP by 3 percent due to a lower interest rate, lessened tax distortion, and reduction in monetary transaction cost. Furthermore, they also find that, in the short-run, the presence of universally accessible interest-bearing CBDC enhances business cycle stability.

The macroeconomic stability implications of CBDC can not be separated from the role of a universally accessible interest-bearing CBDC as a monetary policy tool. In this regard, it delivers a huge implication on monetary policy and enhances the central bank's control over macroeconomic stability. Bordo and Levin (2017) explain that the interest rate embedded in the CBDC provides a secure store of value of money and can serve as the main tool for conducting monetary policy. In this case, the CBDC rate could act as a business cycle stabilizer through which the central bank conducts either the quantity-based or the price-based CBDC policy in a countercyclical manner (Barrdear and Kumhof, 2021;

Bindseil, 2020; Meaning *et al.*, 2021; Nebbi *et al.*, 2022). The CBDC, as a monetary policy instrument, can generate a more effective monetary policy because CBDC is directly connected to every economic agent not only the banking sector but also households and firms (Copic and Franke, 2020; Nebbi *et al.*, 2022). Mancini-Griffoli *et al.* (2018) also suggest that the interest rate channel may be the most affected and could be strengthened by the presence of a universally accessible interest-bearing CBDC through which changes in interest rate directly induce households and firms to rebalance investment and consumption between the future and the present, especially if these are exposed to interest-sensitive borrowing and saving instruments. Using a dataset from Colombia, Nebbi *et al.* (2022) also find that the existence of CBDC enhances macroeconomic stability and effectively reduces the sacrifice ratio – meaning that the output loss due to a reduction in inflation rate is relatively small under the CBDC regime.

While the literature on a universally accessible interest-bearing CBDC has grown significantly in addressing the macroeconomic stability implications of the CBDC, related literature in the context of Indonesia is still relatively limited. This issue is important in the case of Indonesia as Juhro (2021) specifically mentions that Bank Indonesia always pays attention to the potential implications of issuing CBDC on macroeconomic stability. Harahap *et al.* (2017) investigate the role of financial technology improvement embedded in CBDC to explore the impact of CBDC on monetary policy and macroeconomy. Using CGE simulation, their finding suggests find that the issuance of CBDC initially escalates the deposit rate, and it also increases national economic growth and decreases unemployment. Nevertheless, their study addresses the implication of macroeconomic stability. Purnawan and Riyanti (2019) elaborate on the effect of CBDC on the central bank's policy design qualitatively and argue that CBDC provides several benefits such as a new monetary policy instrument, increasing financial inclusion, and reducing transaction costs. Similarly, their study elaborates on the implication of the CBDC on macroeconomic stability. More recently, Zams *et al.* (2020) analyze the optimal CBDC design for Indonesia by collecting experts' opinions. They argue that, based on ANP analysis, non-interest-bearing CBDC is the most ideal type for Indonesia.

III. THE MODEL

Our baseline model is built based on a simple reduced-form NK model. To this end, we first define the supply-side of the economy using the standard form of the forward-backward-looking Lucas supply shock model as follows:

$$\pi_t = \alpha \mathbb{E}_t(\pi_{t+1}) + (1 - \alpha)(\pi_{t-1}) + \lambda y_t + \varepsilon_t^\pi \quad (1)$$

$$\varepsilon_t^\pi = \mu_t^\pi \varepsilon_{t-1}^\pi + \epsilon_t^\pi \quad (2)$$

Where π denotes the inflation rate, y is the output gap, α represents the smoothing coefficient, λ specifies the effect of the output gap on inflation, $\mathbb{E}_t$ is the expectation notation, and ε_t^π expresses the supply shocks with first-order autoregressive form.

Second, we specify the demand side of the economy using the forward-backward-looking IS curve.

$$y_t = \delta \mathbb{E}_t(y_{t+1}) + (1 - \delta)(y_{t-1}) - \tau(i_t - \mathbb{E}_t \pi_{t+1}) + \psi(i_t^{dc} - \mathbb{E}_t \pi_{t+1}) + \varepsilon_t^y \quad (3)$$

$$\varepsilon_t^y = \mu_t^y \varepsilon_{t-1}^y + \epsilon_t^y \quad (4)$$

Where δ denotes the smoothing coefficient of the output gap, i_t is the policy rate, i_t^{dc} represents the CBDC rate, and ε_t^y is the demand shocks expressed by the first-order autoregressive form process. We augment the conventional IS curve expressed in Equation (3) by introducing the effect of CBDC. Different from the policy rate that negatively impacts output, the CBDC rate is expected to influence the output gap positively (Nebbi *et al.*, 2022). The transmission of the changes in the CBDC policy rate to the real economy is reasonably straightforward. Changes in the interest rate embedded in the CBDC influence the income of its holders as follows: The higher (lower) interest rate on CBDC increases the return of holding CBDC and reduces transaction costs. For households, this increases (decreases) household income, and, on the other hand, this further increases (decreases) the ability to consume. Consequently, a higher (lower) interest rate on CBDC makes CBDC more (less) attractive so that it in turn injects (withdraws) liquidity into the economy, expanding (contracting) the demand-side economic activity (Barrdear and Kumhof, 2021; Bordo and Levin, 2017; Gross and Schiller, 2020; Nebbi *et al.*, 2022).

Let us now define the central bank policies aspect. Under the CBDC regime, the central bank conducts a double monetary policy that uses not only the conventional policy rate but also the CBDC rate policy. First, we specify the policy rate reaction function as follows:

$$i_t = \beta_i i_{t-1} + (1 - \beta_i)(\beta_\pi \pi_t + \beta_y y_t) + \varepsilon_t^{icb} \quad (5)$$

$$\varepsilon_t^{icb} = \mu_t^{icb} \varepsilon_{t-1}^{icb} + \epsilon_t^{icb} \quad (6)$$

Where β represents the smoothing coefficient of the policy rate rule and ε_t^{icb} is the monetary policy shock with a first-order autoregressive form.

In this regard, we specify the CBDC rate reaction function as the following equation:

$$i_t^{dc} = \beta_{dc} i_{t-1}^{dc} + (1 - \beta_{dc})(\kappa i_t - (\eta_\pi \pi_t + \eta_y y_t)) + \varepsilon_t^{idc} \quad (7)$$

$$\varepsilon_t^{idc} = \mu_t^{idc} \varepsilon_{t-1}^{idc} + \epsilon_t^{idc} \quad (8)$$

Where $0 < \kappa < 1$ is the fixed spread between the policy rate and the CBDC rate and β_{dc} is the smoothing parameter.

Following Barrdear and Kumhof (2021), the CBDC rate reaction function is specified using a countercyclical rule through which the CBDC rate responds negatively to inflation and the output gap. This suggests that a countercyclical rule of CBDC policy rate works: when the inflation rises (falls), the central bank would respond to lower (higher) the interest rate on CBDC, making it less (more) attractive. In turn, it withdraws (injects) liquidity from the economy and helps to lower (raise) the output and inflation. Consequently, the economic agents convert their CBDC to other risk-free assets and rebalance their consumption and investments.

Our primary analysis relies on a simple procedure in which we compare and evaluate the macroeconomic consequences that arose from the exogenous shocks $\varepsilon_{t+1}^i, i \in \{y, \pi, i^{cb}, i^{dc}\}$ under the non-CBDC and CBDC regimes. Under the CBDC regime, the central bank has two complementing policy instruments and responds to macroeconomic shocks by adjusting the policy rate and the CBDC rate. On the other hand, the central bank only has the policy rate as the monetary policy instrument under the non-CBDC regime. This makes the model of the non-CBDC regime a standard NK three-equation model.

IV. DATA AND CALIBRATION

The parameters of our model are calibrated from related literature and correspond to estimations using the monthly data for Indonesia from July 2005 to February 2022. Our observation is specifically starts at the first time of interest-targeting regime being used by Bank Indonesia and it also covers several economic shocks that affect macroeconomic fluctuation in Indonesia such as the Global Financial Crisis (GFC) in 2008, Taper Tantrum in 2013, and Covid-19 Pandemic in 2020. The data we use to estimate the parameters consists of three observed variables covering the output gap (y), inflation (π), and policy rate (i^{cb}), and the CBDC rate (i^{dc}), as an unobserved variable.

Table 1.
Descriptive Statistics

This table exhibits three observed variables covering the output gap (y), inflation (π), and policy rate (i^{cb}), and the CBDC rate (i^{dc}) throughout July 2005 to February 2022. Source: Authors' calculation from Bank Indonesia and Bank for International Settlements (BIS).

Variable	Observations	Mean	Std. Dev.	Min	Max
y	200	0.00	0.66	-3.54	1.99
π	200	5.62	3.71	1.17	18.38
i^{cb}	200	6.69	2.14	3.50	12.75

Table 1 exhibits the descriptive statistics for the observed variables. The data for the inflation rate is obtained from Bank for International Settlements (BIS) statistics and calculated as the year-on-year percentage change. The data suggests that from July 2005 to February 2022, the average inflation in Indonesia is 5.62 percent. The output gap data, we use Industrial Production Index collected from

Bank Indonesia statistics. More specifically, we calculate the output gap using the proportional gap between actual output and potential output estimated using the Hodrick-Prescott Filter. The data indicates that the output gap in Indonesia varies from -3.45 percent to 1.99 percent. Lastly, we use the BI-7-day Repo rate as the policy rate variable. From November to December 2005, Bank Indonesia set the highest policy of 12.75 percent, while from February 2021 to February 2022 the policy rate reached the lowest rate of 3.50 percent to support post-Covid pandemic economic recovery.

Table 2.
Parameters Calibration

This table shows the parameters for the baseline model. The value of parameters λ , ψ , μ^π , μ^y , μ^{icb} , and μ^{idc} are based on estimations corresponding to our dataset, while other parameters are constrained by the value referred to Fouejieu, Popescu and Villieu (2019) and Barrdear and Kumhof (2021).

Parameters	Value	Reference
α	0.49	Fouejieu, Popescu and Villieu (2019)
λ	2.52	Authors' estimation
δ	0.55	Fouejieu, Popescu and Villieu (2019)
τ	0.99	Fouejieu, Popescu and Villieu (2019)
ψ	1.15	Authors' estimation
β_i	0.98	Fouejieu, Popescu and Villieu (2019)
β_{dc}	0.90	Authors' constraint
β_π	1.50	Fouejieu, Popescu and Villieu (2019)
β_y	0.50	Fouejieu, Popescu and Villieu (2019)
η_π	1.50	Authors' constraint
η_y	0.50	Authors' constraint
κ	0.80	Barrdear and Kumhof (2021)
μ^π	0.86	Authors' estimation
μ^y	0.99	Authors' estimation
μ^{icb}	0.61	Authors' estimation
μ^{idc}	0.85	Authors' estimation

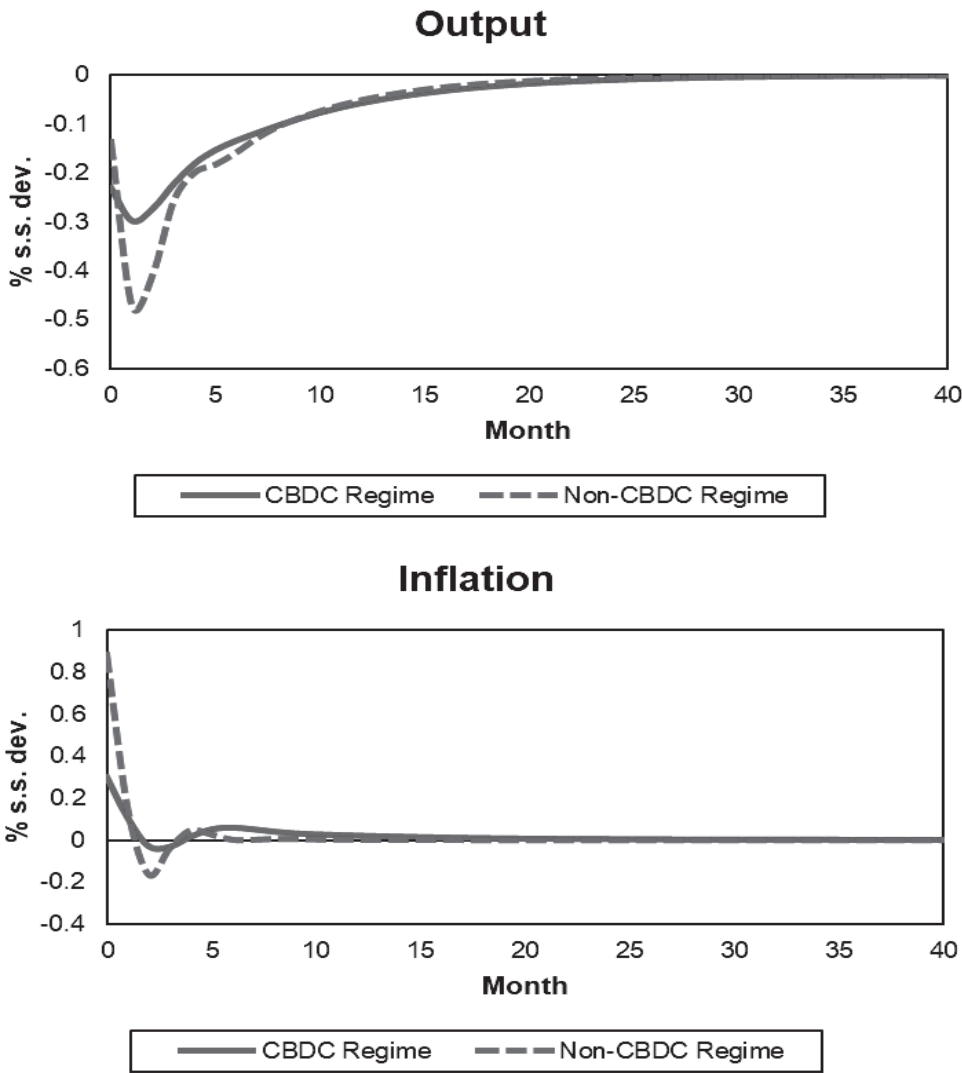
Table 2 shows the value of the parameters we use in the baseline model. For the estimated-based parameters, the estimation obtains $\lambda=2.52$, $\psi=1.15$, $\mu^\pi=0.86$, $\mu^y=0.99$, $\mu^{icb}=0.61$, and $\mu^{idc}=0.85$. Furthermore, the literature-based parameters consist of smoothing parameters $\alpha=0.49$, $\delta=0.55$, $\beta_i=0.98$, and $\beta_{dc}=0.90$, and $\tau=0.99$. The parameter of the fixed spread between the policy rate and the CBDC rate $\kappa=0.80$. More specifically, Barrdear and Kumhof (2021) set no specific numeral of κ but only restrict it to be $0<\kappa<1$.

V. RESULTS AND DISCUSSIONS

Let us first elaborate on our analysis concerning the responses of inflation, output gap, policy rate, and CBDC rate to a unit of supply and demand shocks $\varepsilon^i_{t+1}, i \in \{\pi, y\}$ under the CBDC regime and non-CBDC regime. Generally speaking, the responses of inflation and output to supply and demand shocks are more stable under the CBDC regime than in the non-CBDC regime.

Figure 1.
Response to Supply-side Shocks

Solid blue line represents endogenous variable responses to supply-side shocks under the CBDC regime. The dashed red line denotes endogenous variable responses to supply-side shocks under the non-CBDC regime.



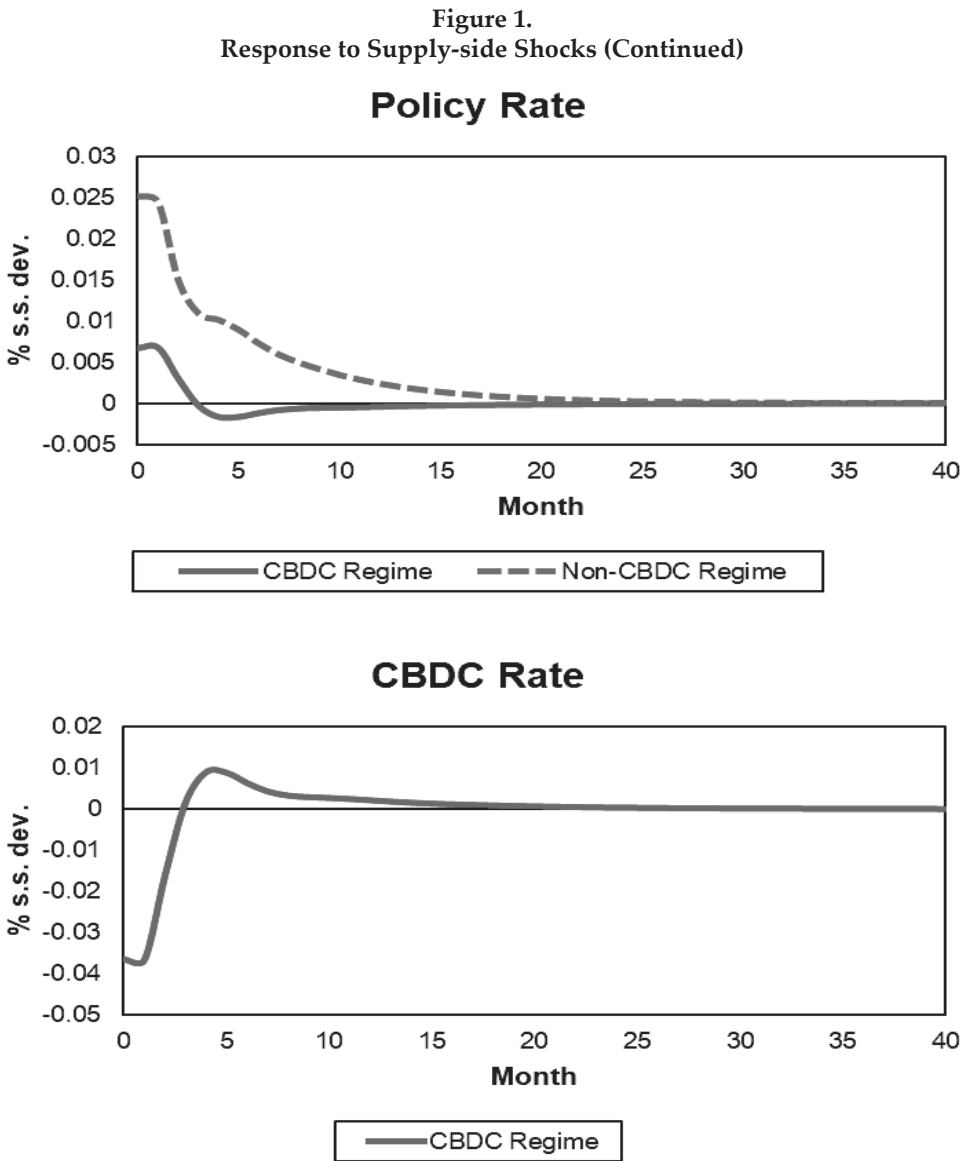


Figure 1 exhibits the responses of macroeconomic indicators and monetary policy due to supply-side shocks under the CBDC regime and the non-CBDC regime. First, the output gap responds to the supply shocks negatively. Under the CBDC regime, the output gap decreases by 0.23 percent at the origin period and by -0.29 percent in the next month. Meanwhile, although the output gap decreases only by 0.13 percent at the origin period, the output gap significantly decreases by 0.4 percent in the next month. Second, the inflation rate increases in response to supply shocks. At the origin month, the inflation rate hikes up to 0.8 percent under

the non-CBDC regime, while it is only a 0.3 percent increase under the CBDC regime. Third, the results show that the policy rate is also less fluctuating under the CBDC regime in which it increases only by 0.01 percent in response to supply shocks. On the other hand, the conventional policy rate surges up to 0.02 percent under the non-CBDC regime. This suggests that the macroeconomic indicators are more stable in response to supply shocks under the CBDC regime than in the non-CBDC regime. We argue that such an enhancement in macroeconomic stability is contributed to the presence of universally accessible interest-bearing CBDC. In this regard, our estimation shows that the CBDC rate, as an additional monetary policy instrument, responsively reacts to supply shocks. More specifically, the CBDC rate decreases by 0.04 percent in response to supply shocks.

Figure 2.
Response to Demand-side Shocks

Solid blue line represents endogenous variable responses to demand-side shocks under the CBDC regime. The dashed red line denotes endogenous variable responses to demand-side shocks under the non-CBDC regime.

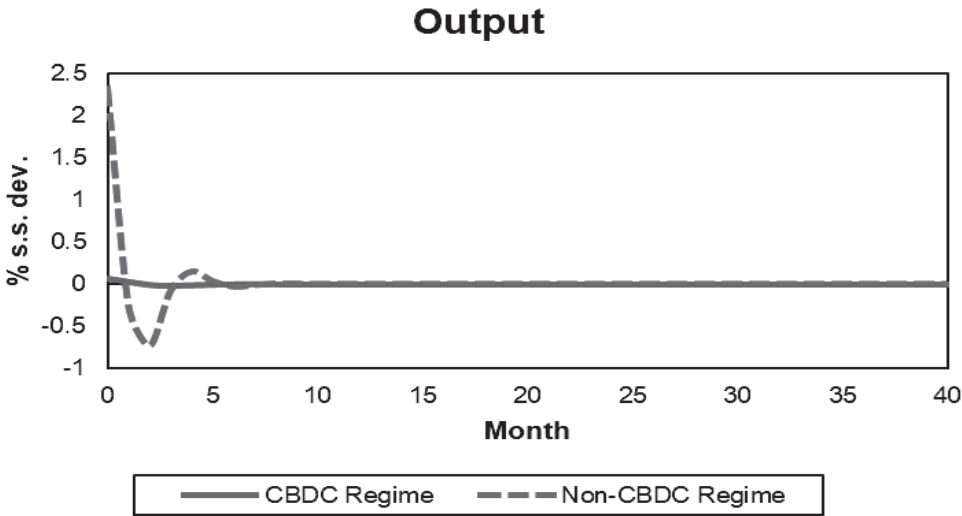


Figure 2.
Response to Demand-side Shocks (Continued)

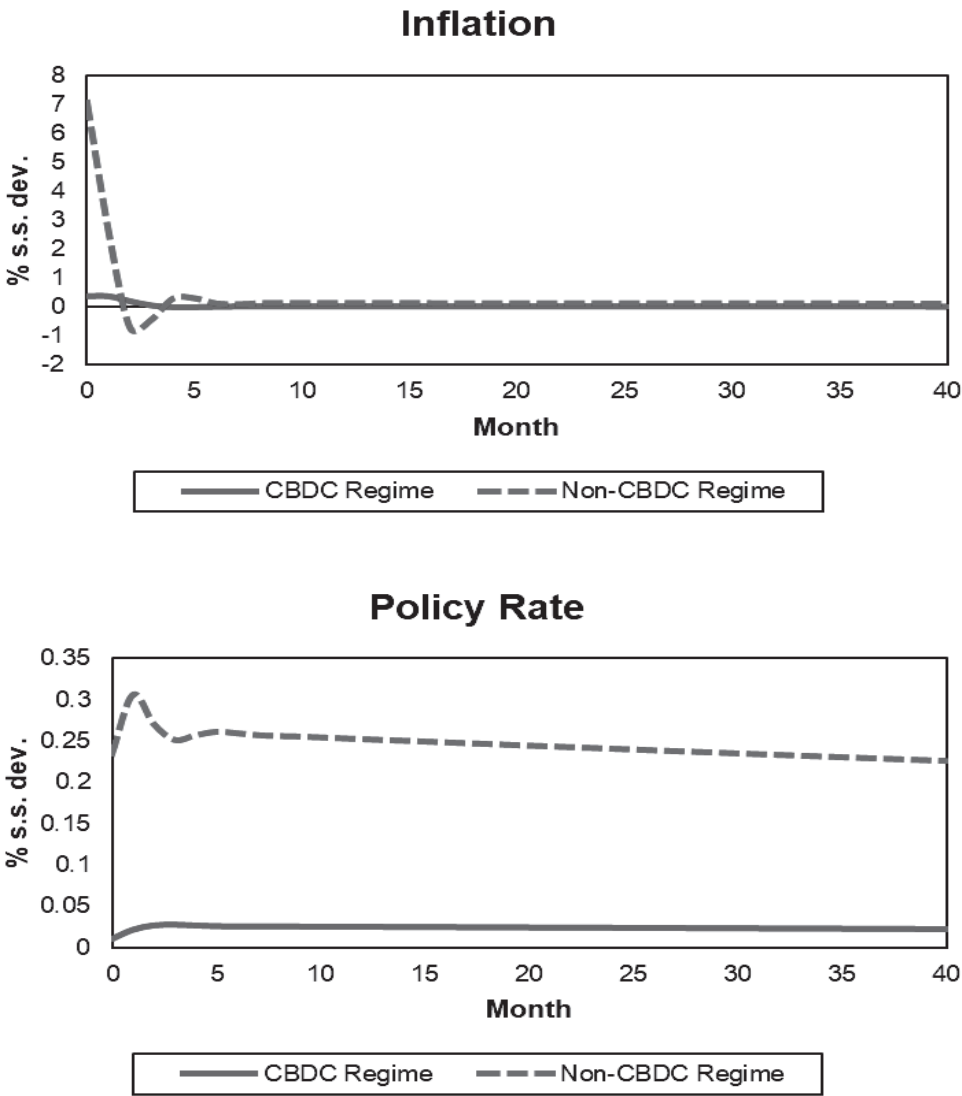
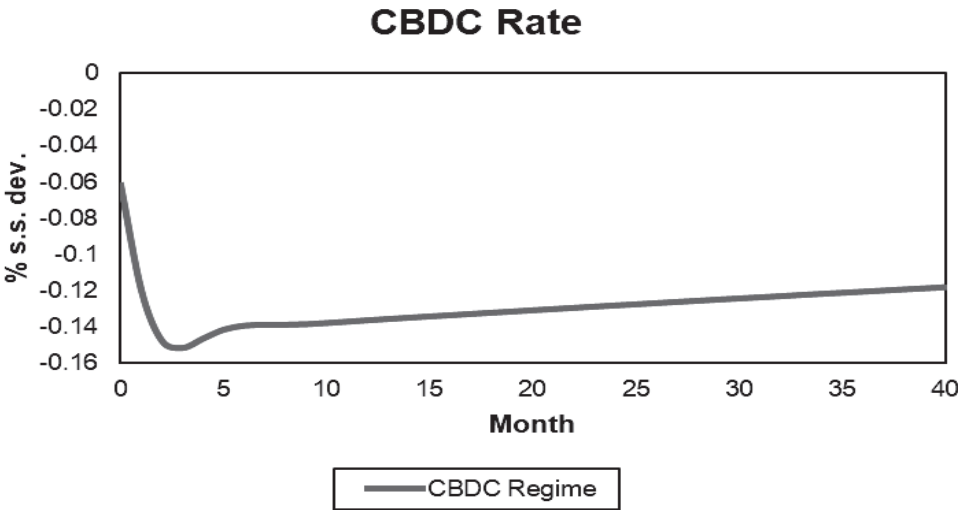


Figure 2.
Response to Demand-side Shocks (Continued)

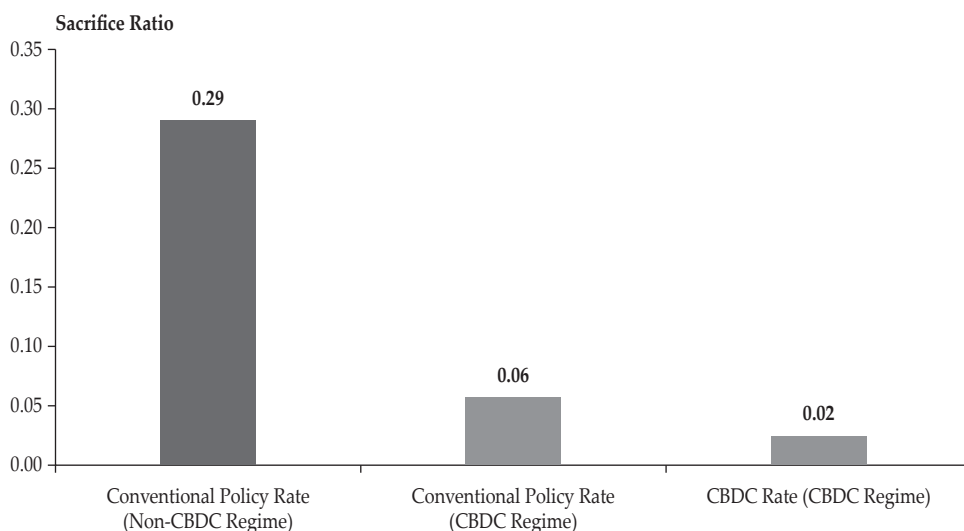


Source: Authors' estimation.

Now we assess the response of macroeconomic indicators and monetary policy in the presence of positive demand shocks. Generally speaking, the output gap and inflation fluctuations in response to demand shocks are more stable under the CBDC regime than in the non-CBDC regime. At the origin month, the results under the CBDC regime indicate that the output gap increases only by 0.7 percent while under the non-CBDC regime it increases by to 2.3 percent. The inflation rate, under the CBDC regime, initially increases by 0.3 percent in response to demand shocks. On the other, the inflation rate surges up to 7 percent under the non-CBDC regime. In this case, the policy rate only responds with a 0.01 percent increase under the CBDC regime, while under the non-CBDC regime it increases by 0.25 percent. For the CBDC rate, it reacts to the demand side shocks by a -0.06 percent. This finding suggests that a countercyclical rule of CBDC policy rate works effectively through which when the inflation rises (falls), the central bank would respond to lower (higher) the interest rate on CBDC, making it less (more) attractive. In turn, it withdraws (injects) liquidity from the economy and helps to lower (raise) the output and inflation (Barrdear and Kumhof, 2021; Meaning *et al.*, 2021; Nebbi *et al.*, 2022).

Figure 3.
Sacrifice Ratio Due to Discretionary Policy Shocks

The vertical axis is the value of the sacrifice ratio, and the horizontal axis is the source of policy shocks comprising the conventional policy rate and the CBDC rate. The red bar represents the sacrifice ratio produced in the non-CBDC regime. The blue bars denote the sacrifice ratio in the CBDC regime. The sacrifice ratio is computed as the slope of the output gap response and inflation responses to the discretionary monetary policy shocks including the conventional policy rate and the CBDC rate.



Source: Authors' estimation.

For the last part of our analysis, we look into the comparative sacrifice ratio produced by the discretionary monetary policy under the CBDC regime and the non-CBDC regime (see Figure 3). More specifically, the sacrifice ratio is computed as the slope of the output gap and inflation responses to the discretionary monetary policy shocks including the conventional policy rate and the CBDC rate (Nebbi *et al.*, 2022). Our analysis suggests that the sacrifice ratio generated by the conventional policy rate under the non-CBDC regime reaches 0.29. In comparison with the CBDC regime, the sacrifice ratio delivered by the conventional monetary policy is only 0.06. This suggests that the effectiveness of the conventional policy rate as monetary policy is significantly enhanced under the CBDC regime. This finding is consistent with Nebbi *et al.* (2022) suggesting that the conventional policy rate is significantly more effective under the CBDC regime. Furthermore, our estimation also finds that the sacrifice ratio produced by the CBDC is even lower than the one produced by the conventional policy rate. We argue that is because the CBDC rate directly affects every economic agent not only the banking sector but also households and firms. This makes the impact of changes in CBDC rate is more directly connected to the real sector under the CBDC regime (Copic and Franke, 2020; Nebbi *et al.*, 2022). Therefore, from these findings, we can conclude that the monetary policy will be more effective under the CBDC regime than in the non-CBDC regime.

VI. ROBUSTNESS CHECKS

To ensure that our primary findings are robust, we employ sensitivity analysis. In this regard, we use alternative model parameters as follows: $\kappa=0.90$, $\kappa=0.70$, $\beta_i=0.90$, and $\beta_i=0.80$.

Figure 4.
Sensitivity Analysis

This figure illustrates the sensitivity analysis. Panel A and Panel B respectively exhibit the response of output gap, inflation, policy rate, and the CBDC rate due to supply-side shocks and demand-side shocks. The parameters we use and compare are $\kappa=0.90$, $\kappa=0.70$, $\beta_i=0.90$, and $\beta_i=0.80$.

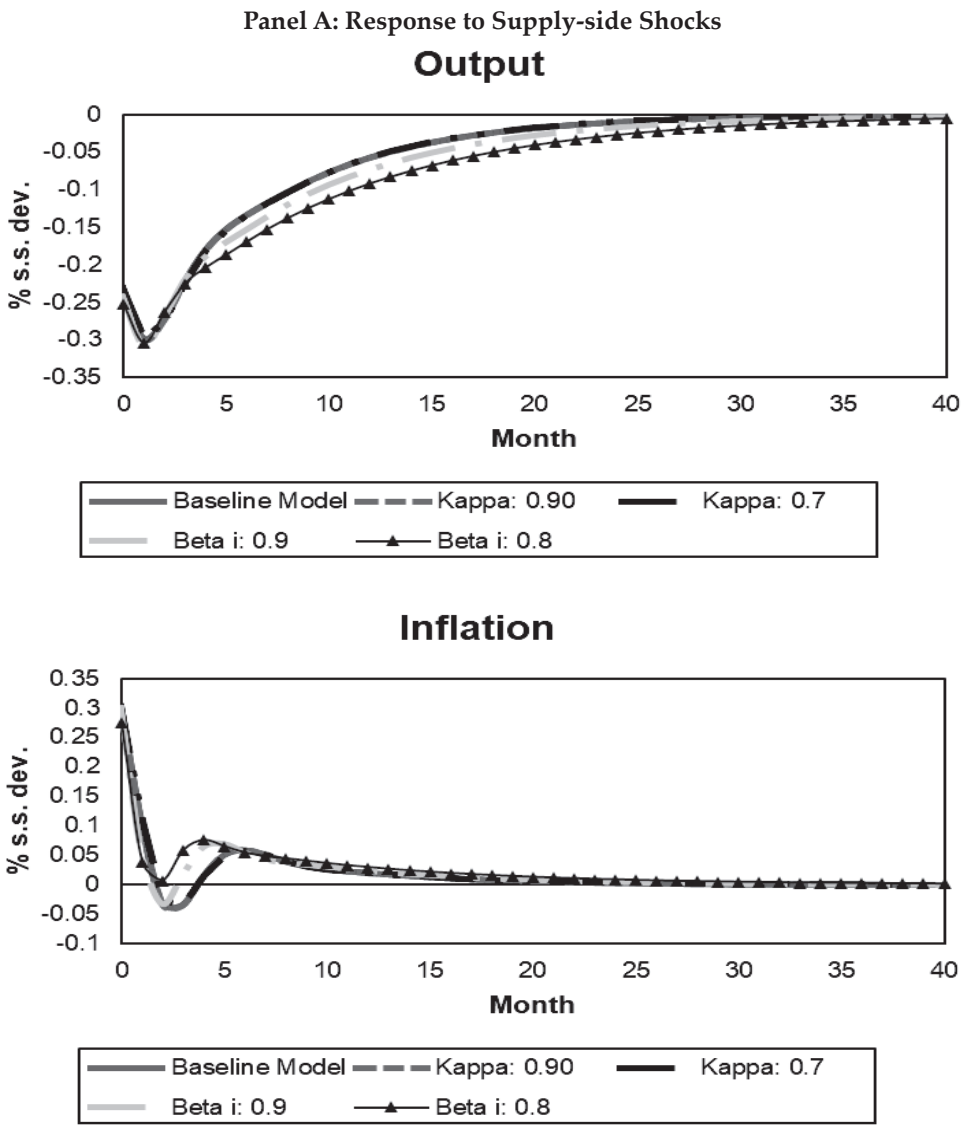


Figure 4.
Sensitivity Analysis (Continued)

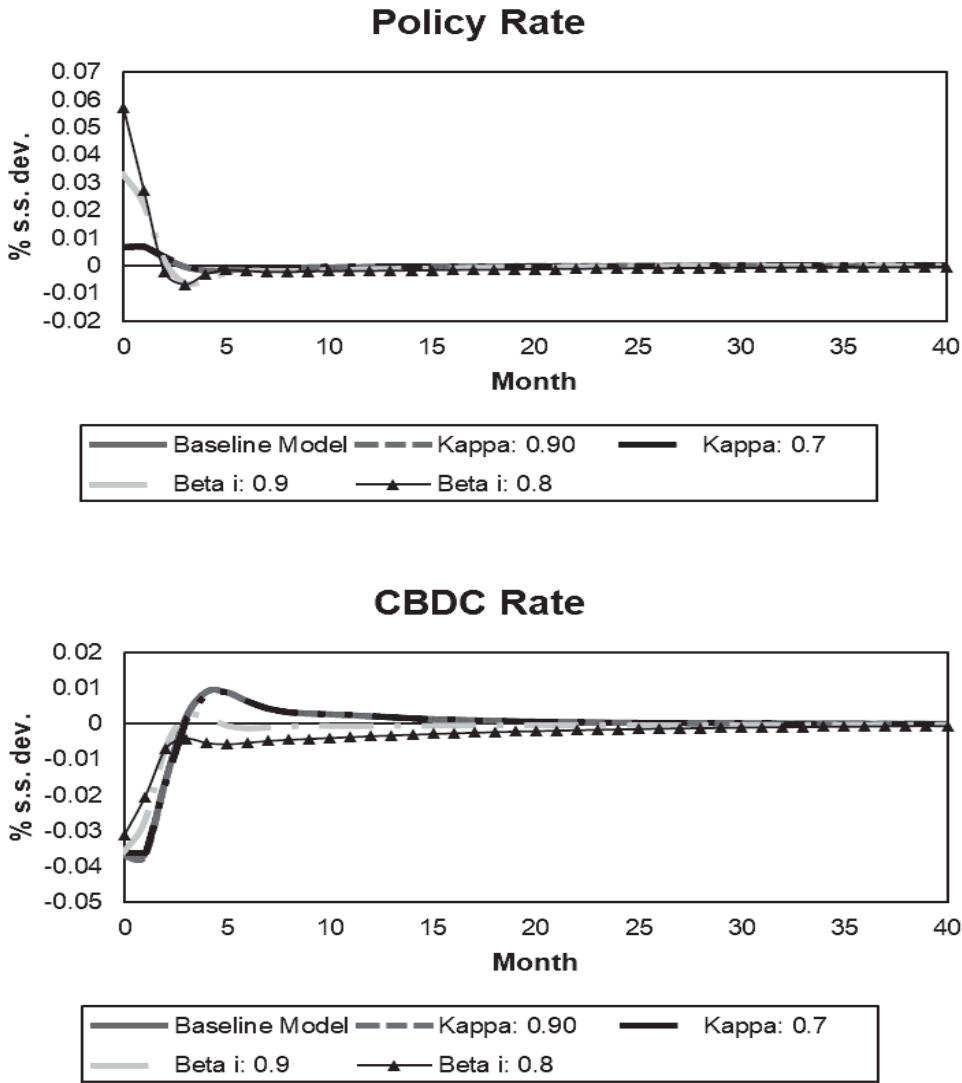
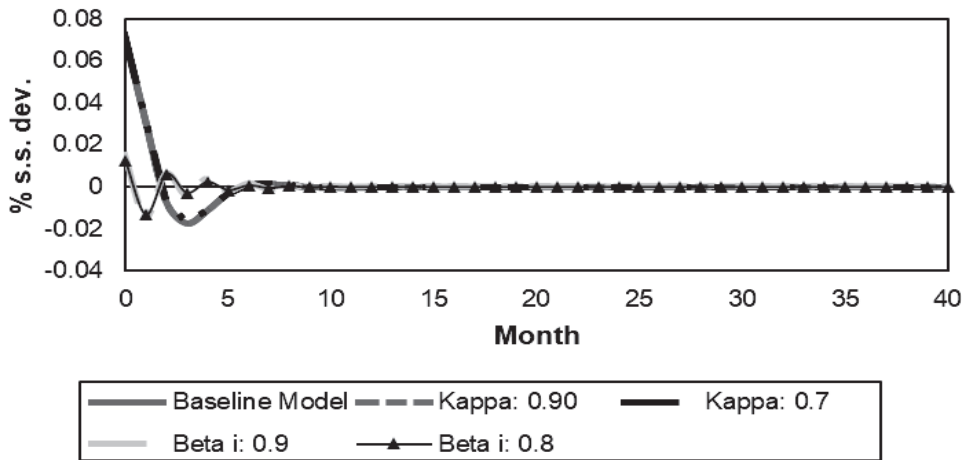


Figure 4.
Sensitivity Analysis (Continued)

Panel B: Response to Demand-side Shocks

Output



Inflation

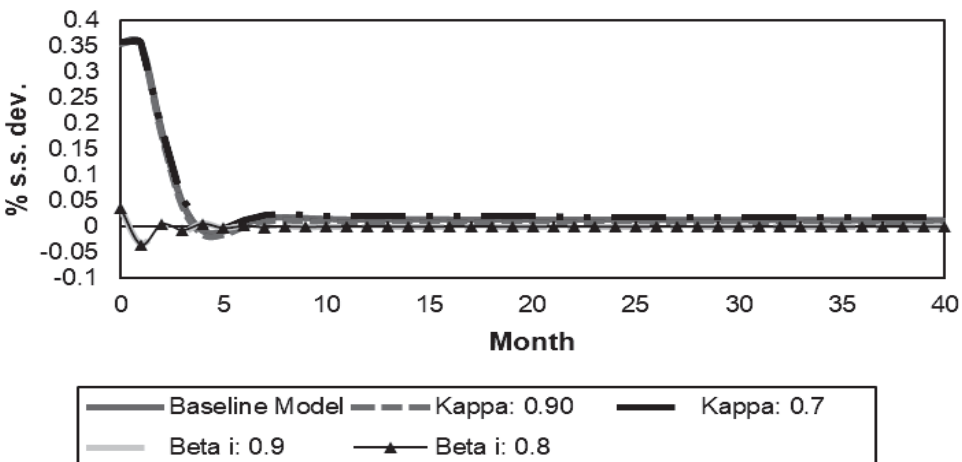


Figure 4.
Sensitivity Analysis (Continued)

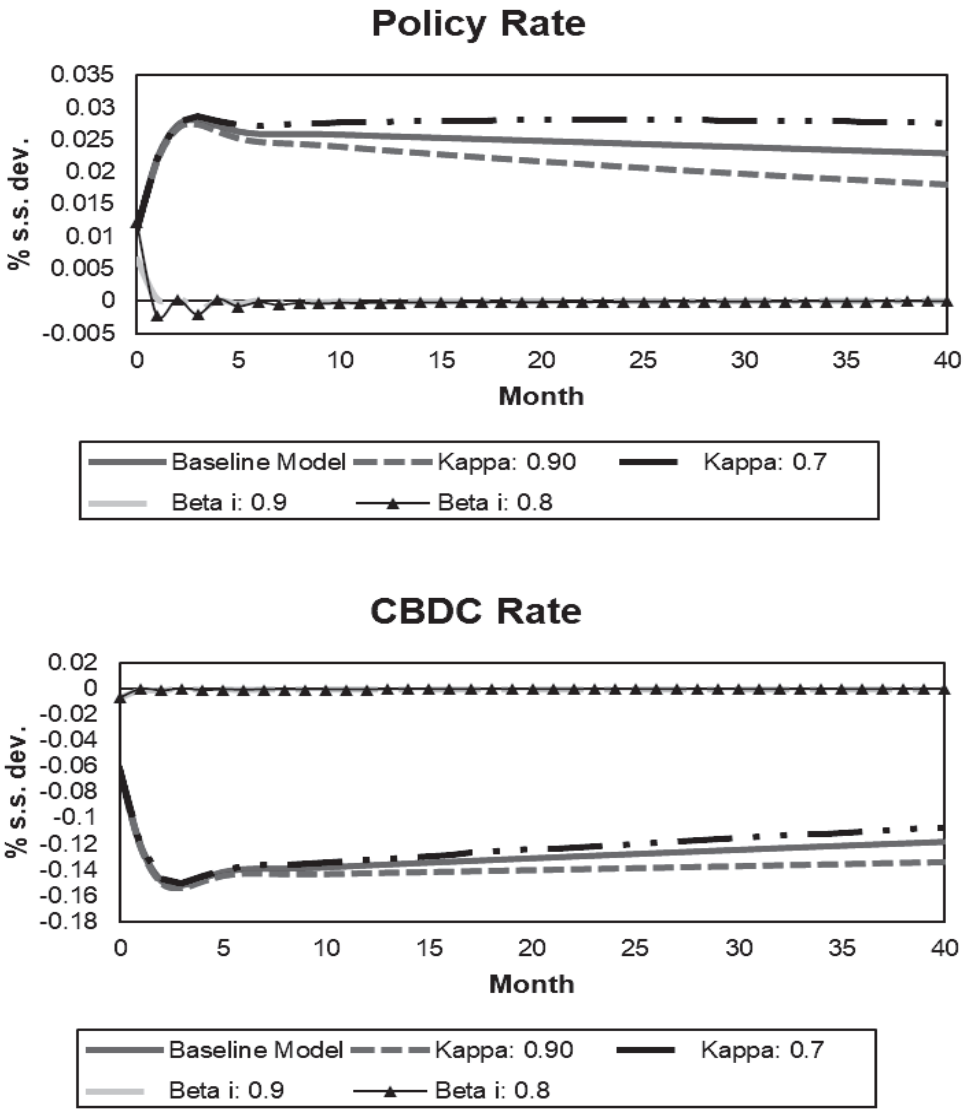
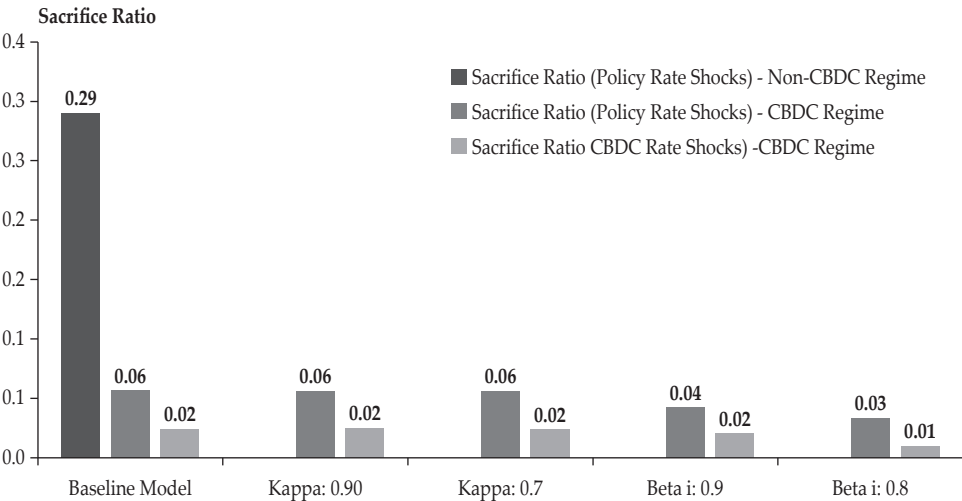


Figure 4 illustrates the impulse response function curves estimated using baseline model parameters and alternative parameters. First, our sensitivity analysis shows that the output gap, inflation, policy rate, and the CBDC rate have nearly identical patterns in response to supply-side shocks (see Panel A). In the case of demand-side shocks, our estimation shows that the response of output gap, inflation, policy rate, and the CBDC is identical across alternative parameters κ , but different when we use alternative smoothing parameters $\beta_i=0.9$ and

$\beta_i=0.8$. Although it is seemingly quite different, the responses of macroeconomic variables and monetary policy due to demand-side shocks are virtually aligned with our hypothesis. More specifically, the CBDC rate decreases by 0.01 percent under alternative β_i model. This, therefore, indicates that our primary findings are robust, especially subject to alternative parameters κ and β_i .

Figure 5.
Sacrifice Ratio Due to Policy Shocks

The vertical axis is the value of the sacrifice ratio and the horizontal axis is the source of policy shocks comprising the conventional policy rate and the CBDC rate. The red bar represents the sacrifice ratio due to the conventional policy rate under the non-CBDC regime. The blue bars denote the sacrifice ratio due to the conventional policy rate under the CBDC regime. Green bars denote the sacrifice ratio due to the CBDC rate. The sacrifice ratio is computed as the slope of the output gap response and inflation responses to the discretionary monetary policy shocks including the conventional policy rate and the CBDC rate. The parameters we use and compare are $\kappa=0.90$, $\kappa=0.70$, $\beta_i=0.90$, and $\beta_i=0.80$.



Source: Authors' estimation.

Lastly, we also estimate the sacrifice ratio using alternative model parameters κ and β_i (see Figure 5). From the results, we can see that the sacrifice ratios generated by the discretionary conventional policy rate under the CBDC regime are lower than that in the non-CBDC, regardless of the parameters we use. Furthermore, our estimation also demonstrates that, under the CBDC regime, the sacrifice ratio produced by the CBDC rate shocks is lower than that of the conventional policy rate. In conclusion, our primary findings regarding the effectiveness of monetary policy are robust.

VII. CONCLUSION

This paper raises several fascinating questions regarding the possible impact of universally accessible and interest-bearing CBDC in Indonesia. The first question that appears is related to whether or not a universally accessible interest-bearing

CBDC supports Bank Indonesia's objective to manage macroeconomic stability. Second, we ask how the interest rate embedded in CBDC, as a policy instrument, will respond to and play a role in stabilizing macroeconomic fluctuations. Lastly, we also look into whether the existence of CBDC enhances the effectiveness of the monetary policy.

To address our research questions, we first analyze the magnitude of macroeconomic fluctuations in response to the demand and supply shocks under the CBDC regime and non-CBDC regime. Second, we scrutinize the role of the interest rate embedded in CBDC as a policy instrument in response to macroeconomic shocks. Third, we investigate the effectiveness of monetary policy in the CBDC regime and the non-CBDC regime. For this end, we construct a simple reduced-form New Keynesian (NK) model calibrated to Indonesia's monthly data. We characterize the existence of a universally accessible interest-bearing CBDC by introducing the CBDC policy rate which is conducted separately from the conventional policy rate. In this regard, we specify the CBDC rate reaction function using a countercyclical rule in a tradition of Barrdear and Kumhof (2021).

This paper contributes to the current literature by highlighting fresh insights for Indonesia. We find that the CBDC regime significantly improves macroeconomic stability in Indonesia. Our result indicates that macroeconomic shocks are significantly less volatile and more quickly stabilized under the CBDC regime than under the non-CBDC regime. In this regard, the CBDC rate, as an additional monetary policy instrument, responsively reacts to macroeconomic fluctuations and significantly impacts the demand side. Our results also suggest that the sacrifice ratio, as the proxy of monetary policy effectiveness, produced under the CBDC regime due to discretionary policy shocks is significantly lower than in the non-CBDC regime – meaning that the output loss due to a reduction in the inflation rate is relatively small under the CBDC regime. This indicates that monetary policy is more effective under the CBDC regime than in the non-CBDC. Our findings are robust to sensitivity checks using the alternative value of several key parameters.

Based on our results, we suggest that policymakers, especially Bank Indonesia, need to consider issuing a universally accessible interest-bearing CBDC in Indonesia. Our results show that it brings several economic benefits, from macroeconomic stability to a lower degree of sacrifice ratio. However, this requires robust preparation, especially related to the technical aspect, cyber security aspect, and political economy aspect. Departing from this concern, Bank Indonesia needs to work out the long-run master plan before issuing CBDC in Indonesia.

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